

Homework 2 (Due April 29, 2020, 23:00)

BLM 2502: Theory of Computation — Spring 2020

Print family (or last) name: _____

Print given (or first) name: _____

Print given student number: _____

I see that this homework has 10 questions in total 5 pages.

I agree that I have to submit my homework solution before the deadline (April 29, 2020, 23:59⁰⁰) otherwise my homework solution will not be accepted and will not be graded. I accept that ***I will add the signed version of this instruction page as a first page into my homework solution;*** otherwise my homework solution will not be graded. I know that ***I have to give my solutions written on white A4-sized pages;*** otherwise my homework solution will not be graded. I will take care of the readability of my solutions, from which I may lose 10 points. For any proofs, I am sure to provide a step-by-step argument, with justifications for every step. I understand that, during solving this homework, it is prohibited to exchange information about solutions with any other person in any way, including by talking or ex-changing solutions / papers.

I know that the course book is “Introduction to the Theory of Computation, 2nd Ed., Massachusetts Institute of Technology, by Micheal Sipser.”

I have read, understand and accept all of the instructions above. On my honor, I pledge that I have not violated the provisions of the Academic Integrity Code of Yıldız Technical University.

Signature and Date

1	2	3	4	5	6	7	8	9	10
20 pts	20 pts	20 pts	20 pts	20 pts	20 pts	20 pts	20 pts	20 pts	20 pts

Total
200 pts

1. **{20 points}** For each of the following languages, state the class of the language (whether it is regular, context-free (but not regular) or neither). Prove your answer. Make sure, if you claim that a language is context free, that you show that it is not also regular.

(a) $L = \{w \in \{0,1\}^* : \exists k \geq 0 \text{ and } w \text{ is a binary encoding (leading zeros allowed) of } 2k+1\}$.

(b) $L = \{a^*b^*c^* - \{a^n b^n c^n : n \geq 0\}\}$.

(c) $L = \{x \in \{a, b\}^* : |x| \text{ is even and the first half of } x \text{ has one more } a \text{ than the second half}\}.$

2. ~~20 points~~ Let $L = \{w \in \{a, b\}^* : \text{the first, middle, and last characters of } w \text{ are identical}\}.$

a. Show a context-free grammar that generates L .

b. Design a PDA that accepts L .

- c. Prove that L is not regular.

3. ~~{20 points}~~ CC Consider the following grammar G :

$$S \rightarrow 1 S 1 \mid T$$

$$T \rightarrow 1 X 1 \mid X$$

$$X \rightarrow 0 X 0 \mid 1$$

- a. What are the first (shortest) four strings $L(G)$?
- b. Give an example of a string $w \in \{0, 1\}^*$ such that $|w| > 7$ and $w \notin L(G)$.

c. Show that G is ambiguous.

4. Find grammars for the given languages

a. $L_1 = L(aaa^*b + b)$.

b. $L_2 = \{a^n c^m b^n : m, n \geq 1\}$.

c. $L_3 = \{a^n b^{n+1} : n \geq 0\}$

5. **[20 points]** Convert the [Grammars-grammars](#) given below to Chomsky Normal Form ~~of the~~. Do not forget to give modified 4-Tuple Grammar [\(The formal definition, or the tuple that has the list of productions, start symbol, etc\).](#)

a. $S \rightarrow a \mid aA \mid B \mid C$
 $A \rightarrow aB \mid \varepsilon$
 $B \rightarrow Aa$
 $C \rightarrow bCD$
 $D \rightarrow bbb$

- b. $S \rightarrow aA|aBB$
 $A \rightarrow aaA|\epsilon$
 $B \rightarrow bB|bbC$
 $C \rightarrow B$

6. ~~{20 points}~~ Find a context-free grammar that generates the language accepted by the PDA

$M = (\{q_0, q_1\}, \{a, b\}, \{A, z\}, \delta, q_0, \{q_1\})$, with transitions

$\delta(q_0, a, z) = \{(q_0, Az)\},$

$\delta(q_0, b, A) = \{(q_0, AA)\},$

$\delta(q_0, a, A) = \{(q_1, \epsilon)\}$

7. ~~{20 points}~~ Determine whether the following languages are context-free or not

a. $L = \{a^n w w^R a^n : n \geq 0, w \in \{a, b\}^*\}$

b. $L = \{a^n b^j a^n b^j : n \geq 0, j \geq 0\}$.

c. $L = \{a^n b^j a^j b^n : n \geq 0, j \geq 0\}$.

d. $L = \{a^n b^j a^k b^l : n + j \leq k + l\}$.

e. $L = \{a^n b^j a^k b^l : n \leq k, j \leq l\}$.

f. $L = \{a^n b^n c^j : n \leq j\}.$

g. $L = \{w \in L((a + b + c)^*) : n_a(w) = n_b(w) = 2n_c(w)\},$ where $n_x(w)$ denotes the number of symbol X in the string w .

8. ~~20 points~~ Let L be a context-free language. Prove that there exists an integer $p \geq 1$, such that the following is true:

For every string s in L with $|s| \geq p$, there exists a string z in L such that $|s| < |z| \leq |s| + p$.

- | 9. ~~{20 points}~~ For the context free languages **in problem 7**, find the grammar of the language in Chomsky Normal Form.

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- | 10. ~~{20 points}~~ For the grammars **in problem 9**, obtain the PDA using the obtained context free grammar.

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